Benjamin Ekow Attabra

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Education

Kwame Nkrumah University of Science & Technology **Ghana**

BSc. Telecommunication Engineering, Second Class Upper *Jan 2023 – Sep 2026*

Relevant Coursework: Linear Algebra, Calculus 1–3, Programming I/II (Python & MATLAB), Data Science, Signals & Systems, Microprocessors & Digital Systems.

Honors & Awards: YARA Research Fellow 2025, Claude Campus Ambassador 2026

Skills & Interests

Skills: Python, Pandas, Numpy, Matplotlib, Scikit-Learn, Jupyter, Tensorflow, and MATLAB

Certifications: Data Analytics (Quantium – Forage), Quantitative Research & Investment Banking (JP Morgan – Forage), and Ethical Hacking Essentials (Virtual InfoSec)

Professional Experience

Young Africans Research Academy

Research Fellow *May 2025 – Present*

- Selected as 1 of 10 fellows from 600+ applicants for a competitive pan-African research fellowship.
- Completed training in research design, literature review, data collection, quantitative & qualitative analysis, and academic writing.
- Preparing independent research on *Sequential Deep Learning for Credit Risk Modelling in Data-Constrained Environments*, with emphasis on financial inclusion and applied AI.

Claude Builder Club @ KNUST

Co-Founder & Co-President *Jan 2025 – Present*

- Co-managed fiscal allocation for API procurement and computational resources for LLM development.
- Directed deployment of Claude-based prompt engineering frameworks and evaluation benchmarks.
- Coordinated the technical resources and sandbox logistics for club-wide AI hackathons.

IEEE KNUST ComSoc Student Branch

Treasurer *Aug 2025 – Present*

- Managed and reconciled all Branch finances and accounts, ensuring strict adherence to IEEE compliance policies.
- Chaired the Fundraising Committee, generating project revenue and securing external sponsorships to support Branch operations.
- Directed strategic budget allocation across the Executive Committee for key events and project initiatives.

Ghana Engineering Students Association

Trade & Tech Committee Deputy Chair *Sep 2025 – Present*

- Appointed under the GESA Constitution to co-lead the TRATECH Committee, tasked with promoting trade, technology, and innovation development among engineering students.
- Assist in reviewing, refining, and implementing innovative student projects, fostering entrepreneurship and applied research across the College of Engineering.
- Lead Tutor, GESA Digital School: designed practical curricula for digital skills; leading Programming, Machine Learning, and Cybersecurity programs.

NASA Space Apps Challenge – Kumasi Local Event

Volunteer *Oct 2025*

- Provided tech support for 20+ participants, troubleshooting issues, to ensure seamless operation of the 48-hour hackathon.
- Coordinated pre-event logistics and physical setup, efficiently deploying essential resources (power, internet access, and workstations) for 5+ competing teams.

Relevant Projects

Sequential Deep Learning for Credit Risk Modeling: GitHub
a credit-risk prediction project built around temporal feature engineering and sequential deep learning on mobile money transaction histories (per-user sequences). Includes multiple baseline and sequence models (e.g., LR/XGB/RF/LightGBM plus LSTM/Hybrid LSTM), with tooling for 5-fold CV benchmarking, feature ablations, and hyperparameter tuning.

EdgeTutor: GitHub
an offline, on-device RAG tutoring assistant for Android, aimed at helping first-year engineering students (KNUST) query textbooks without internet. Supports PDF/TXT ingestion, retrieval + answering locally, and is designed to run on ~1 GB RAM devices using lightweight/quantized models (notably Gemma-3-270M) plus an ONNX embedder pipeline. Includes both a Python MVP prototype and an Android (Kotlin/Jetpack Compose) app.

OFDM-UNet: GitHub
a compact 2D U-Net for time–frequency interference suppression in OFDM. Operates on the OFDM resource grid (Re/Im channels) and predicts a single real-valued mask ($M[t,k] \in [0,1]$) applied element-wise to attenuate interference, trained with a weighted complex-L1 loss (with extra weight on pilot bins). Includes a full pipeline: OFDM preprocessing/dataset generation, training CLI, baseline comparisons, evaluation methodology (EVM/BER), tests, and a demo notebook.

References

Available Upon Request